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13. Write a pandas program to detect missing values of given dataframe

Aim

To write a Pandas program that detects missing values in a DataFrame and displays the result as **True** (missing value) or **False** (no missing value).

Procedure

1. Import the **Pandas** library and read the CSV file into a DataFrame.
2. Display the original DataFrame to view the available data.
3. Use the `isnull()` function to check each cell for missing values.
4. Display the resulting DataFrame containing **True** and **False** values, where **True** indicates a missing value and **False** indicates a valid value.
5. Verify the output and terminate the program.



```
13th.py - C:/Users/hp/AppData/Local/Programs/Python/Python313/13th.py (3.13.9)
File Edit Format Run Options Window Help
import pandas as pd

df = pd.read_csv("C:/Users/hp/Downloads/13th.csv")

print("Original DataFrame:")
print(df)

print("\nMissing Values (True/False):")
print(df.isnull())
```

OUTPUT:

```

IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
----- RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python313/13th.py -----
Original DataFrame:
   ord_no  purch_amt  ord_date  customer_id  salesman_id
0  70001.0    150.50  2012-10-05         3002         5002.0
1      NaN    270.65  2012-09-10         3001         5003.0
2  70002.0     65.26      NaN         3001         5001.0
3  70004.0    110.50  2012-08-17         3003         NaN
4      NaN    948.50  2012-09-10         3002         5002.0
5  70005.0   2400.60  2012-07-27         3001         5001.0
6      NaN    5760.00  2012-09-10         3001         5001.0
7  70010.0   1983.43  2012-10-10         3004         NaN
8  70003.0   2480.40  2012-10-10         3003         5003.0
9  70012.0    250.45  2012-06-27         3002         5002.0
10      NaN     75.29  2012-08-17         3001         5003.0
11 70013.0   3045.60  2012-04-25         3001         NaN

Missing Values (True/False):
   ord_no  purch_amt  ord_date  customer_id  salesman_id
0  False     False     False     False     False
1    True     False     False     False     False
2  False     False     True     False     False
3  False     False     False     False     True
4    True     False     False     False     False
5  False     False     False     False     False
6  False     False     False     False     False
7  False     False     False     False     True
8  False     False     False     False     False
9  False     False     False     False     False
10   True     False     False     False     False
11  False     False     False     False     True

```

15.Aim

To write a Pandas program to identify and replace missing values in a DataFrame with appropriate values so that the dataset contains complete and meaningful information.

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1. Import the Pandas library and read the CSV file into a DataFrame.
2. Display the original DataFrame containing missing values.
3. Replace invalid or missing entries such as ?, --, and blank values with NaN.
4. Fill the missing values using a suitable replacement value such as "Not Available".
5. Display the updated DataFrame after replacing the missing values.

CODE:

```
14th.py - C:/Users/hp/AppData/Local/Programs/Python/Python313/14th.py (3.13.9)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np

df = pd.read_csv("C:/Users/hp/Downloads/14th.csv")

df.replace(["?", "--", ""], np.nan, inplace=True)

df.fillna("Not Available", inplace=True)

print("Updated DataFrame:")
print(df)
```

OUTPUT:

```
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
----- RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python313/14th.py -----
Updated DataFrame:
   ord no  purch amt  ord_date  customer_id  salesman_id
0    70001      150.5  Not Available      3002         5002
1  Not Available    270.65  2012-09-10      3001         5003
2    70002      65.26  Not Available      3001  Not Available
3    70004      110.5  2012-08-17      3003         5001
4  Not Available    948.5  2012-09-10      3002  Not Available
5    70005     2400.6  2012-07-27      3001         5002
6  Not Available     5760  2012-09-10      3001         5001
7    70010  Not Available  2012-10-10      3004  Not Available
8    70003      12.43  2012-10-10  Not Available         5003
9    70012     2480.4  2012-06-27      3002         5002
10  Not Available    250.45  2012-08-17      3001         5003
11    70013     3045.6  2012-04-25      3001  Not Available
>>>
```